SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Project Initialization and Planning Phase

Date	2July2026
Team ID	----------------
Project Title	SmartLender - Applicant Credibility Prediction for Loan Approval
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The **Smart Lender – Loan Eligibility Prediction** project aims to transform the traditional loan approval process by using machine learning algorithms to predict the eligibility of loan applicants. The proposed system improves the speed, accuracy, and consistency of loan approval decisions by analyzing applicant information such as income, employment status, education, loan amount, loan term, and credit history. The solution reduces manual effort, minimizes human bias, and supports financial institutions in making reliable lending decisions through a user-friendly Flask web application.

Project Overview	
Objective	The primary objective is to revolutionize the loan approval process by implementing advanced machine learning techniques, ensuring faster and more accurate assessments.
Scope	The project focuses on collecting and preprocessing loan applicant data, training multiple machine learning models (Decision Tree, Random Forest, K-Nearest Neighbors, and XGBoost), selecting the best-performing model, integrating it with a Flask web application, and deploying the solution on IBM Cloud for real-time loan eligibility prediction.
Problem Statement	
Description	Traditional loan approval processes rely heavily on manual verification and evaluation, making them time-consuming, inconsistent, and prone to human errors. Financial institutions require a reliable system that can quickly analyze applicant information and accurately predict loan eligibility.

Impact	The proposed system improves operational efficiency, reduces processing time, minimizes human bias, enhances prediction accuracy, and helps financial institutions make consistent and reliable lending decisions while improving customer satisfaction.
Proposed Solution	
Approach	The project applies machine learning algorithms to analyze applicant information and predict loan eligibility. After comparing the performance of Decision Tree, Random Forest, KNN, and XGBoost models, the best-performing model is integrated into a Flask web application for real-time prediction.
Key Features	• Machine learning-based loan eligibility prediction. • Applicant-friendly web interface developed using Flask. • Real-time loan approval/rejection prediction. • High prediction accuracy using the XGBoost model. • Faster and more consistent decision-making. • Deployment on IBM Cloud.

	- Real-time decision-making for quicker loan approvals. - Continuous learning to adapt to evolving financial landscapes.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Processor	Intel Core i3 or above
Memory	RAM	8 GB

Storage	Disk space	10 GB Free Storage
Software		
Frameworks	Python frameworks	Flask
Libraries	ML & Data Science Libraries	Pandas, NumPy, Scikit-learn, XGBoost, Matplotlib, Seaborn, Pickle
Development Environment	IDE	Jupyter Notebook, VS Code / PyCharm, Anaconda Navigator
Data		
Dataset	Loan Eligibility Dataset	CSV format (Google Drive dataset provided in SkillWallet / Loan Prediction Dataset)